



P096: White Light Interferometry

Abhishek Iyer (112401002)
Hemanth Kashyap Gundamaraju (112301009)



Objectives

- To simulate White Light Interferometer by using randomly generated CSV file of heights and generating interferograms that might form using the resulting heightmap.
- To create a functioning White Light Interferometer and use AI and Deep Learning algorithms to convert the interference waveform obtained from the interferometer to a 3-D Surface Height Map of the sample surface.

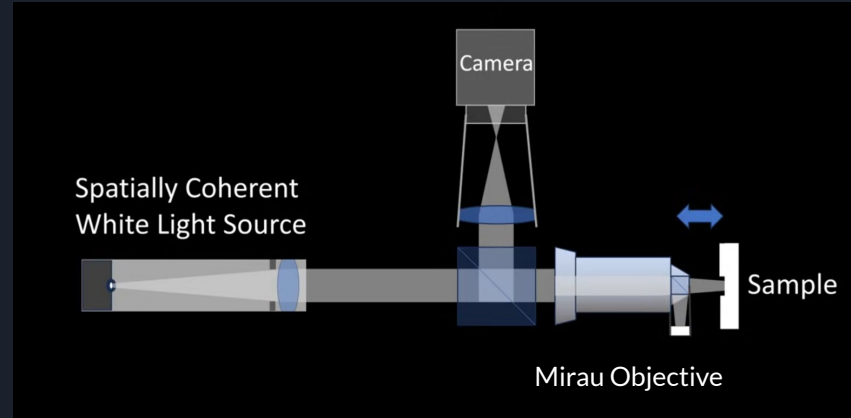


Timeline / Steps

1. Create a program to generate arbitrary height maps
2. Create a program to convert those height maps into corresponding interference pattern
3. Using 1 and 2, create a large dataset for training the AI Model
4. Optimize and fine tune the model until it can generate the corresponding height map when fed input of an interference waveform
5. Create a prototype White Light Interferometer to generate interference waveforms for a given sample surface
6. Further refine the Model to account for any imperfections such as dust/camera aberrations etc. in real world usage
7. Using the AI Model, find out the height variations and surface topography of the sample surface

Working Principle

- White light has a short coherence length, allowing it to generate clear, localized fringes only over a small depth of focus.
- We use a pinhole along with a lens to make the white light spatially coherent.
- The variations in path length caused by the surface topography of the sample surface cause interference between the reflected and reference beams of light, giving rise to interference patterns.





Underlying Physics

- Height variations in the sample surface lead to an interference pattern being generated due to phase differences arising from path difference of the reflected and reference light beams.
- The sample surface is scanned across, generating many fringes
- At each point, the position of maximum fringe contrast (coherence peak) is recorded. This indicates the point where the two beams have equal optical path length.
- The vertical position of this coherence peak gives us the height of the sample at that point

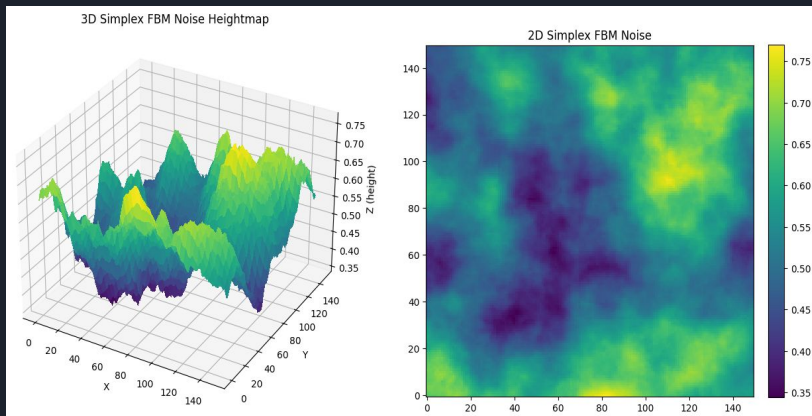


Equipment / Materials Needed

1. Mirau / Michelson Objective Lens
2. Beam Splitter
3. Broad Band Emitter (White LED)
4. Achromatic Lens (to focus the camera onto the sample)
5. Optical Rails + Mounts (Very precise)
6. CMOS Camera
7. Piezo Activator (To scan along the sample)

Step 1: Heightmap generation

- Tested various noise generators:
 - Worley Noise: Too regular
 - Perlin Noise: Worked well
 - Simplex FBM Noise: Improved on efficiency and randomness over Perlin
- Created a Python program to generate random, arbitrary height maps using Simplex Noise Fractal Brownian Motion (Simplex FBM)



- The left image is the heightmap visualized in 3D
- The right image is the 2D projection of the heightmap with colours representing depth



Step 2: Creating a WLI Simulator

- Imported the heightmap from CSV.
- Scaled the data into nanometres for optical relevance.
- Defined simulation parameters: wavelength (λ) 600 nm, coherence length (L_c) 100 nm, and mirror scan range (s) -300 to 300 nm.
- Computed the optical path difference (OPD) using: $OPD=2(z-s)$
- Generated intensity values using cosine interference with a Gaussian coherence envelope.

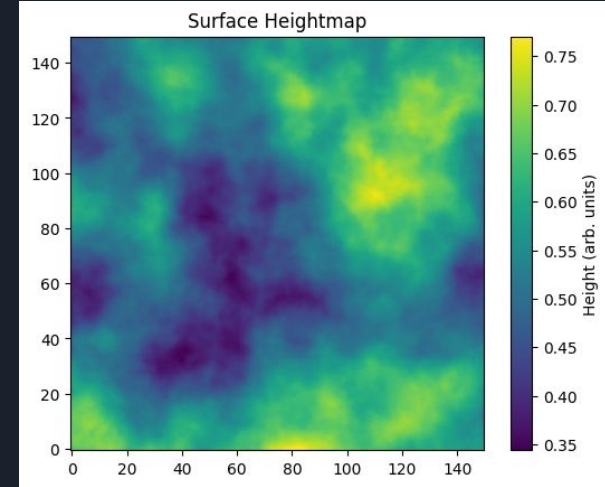
$$I = 1 + e^{-\frac{OPD^2}{2L_c^2}} \cos\left(\frac{2\pi}{\lambda} * OPD\right)$$

Input Surface Data & Heightmap Visualization

- We plotted a 2D heightmap loaded from heightmap.csv to verify the imported data is correct before simulation.

Key takeaways:

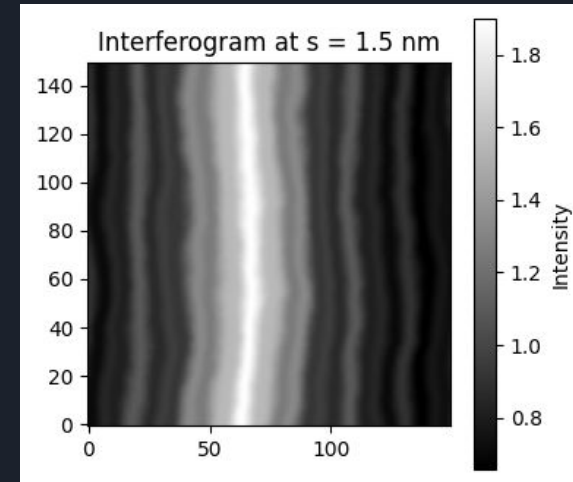
- Colour corresponds to height variation.
- Surface features influence the phase and fringe structure in later steps.
- Ensures the dataset is correctly structured and scaled.



Simulated Interferogram at Fixed Reference Position

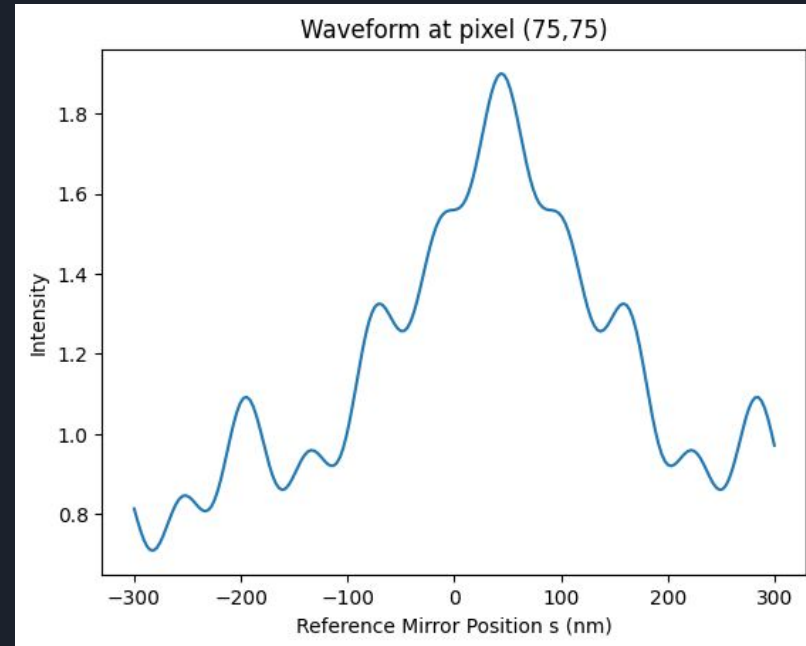
Plot shown:

- One frame of the simulated interferogram: $I[:, :, \text{mid_index}]$
- Shows how interference appears for one mirror position in the scan.
- Areas with varying height show local changes in fringe spacing.
- Visible fringes correspond to areas where the OPD is close to a multiple of the wavelength.
- Fringe contrast is highest near the coherence peak.
- Structure of fringes reflects underlying surface gradients.



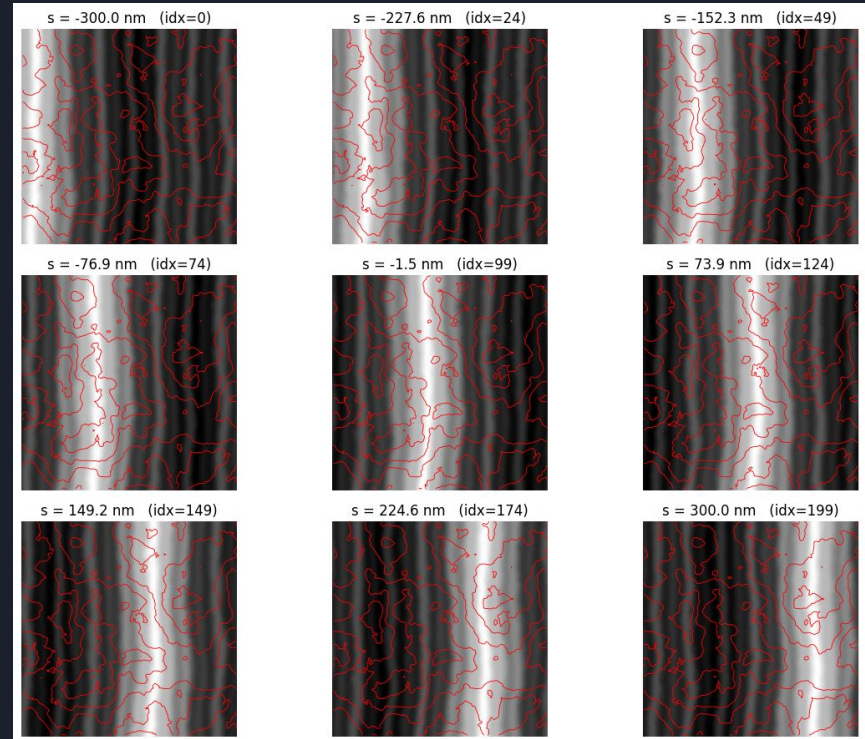
Pixel-Level Behaviour

- Produces a sinusoidal signal modulated by a Gaussian-like envelope.
- The location of maximum envelope intensity corresponds to the surface height at that pixel.
- Used later for height reconstruction.



Global Trends

- Mean intensity vs scan position confirms expected coherence peak behaviour.
- Frames across the scan show fringes appearing, strengthening, then fading.
- The simulation output aligns with expected white-light interferometry response.



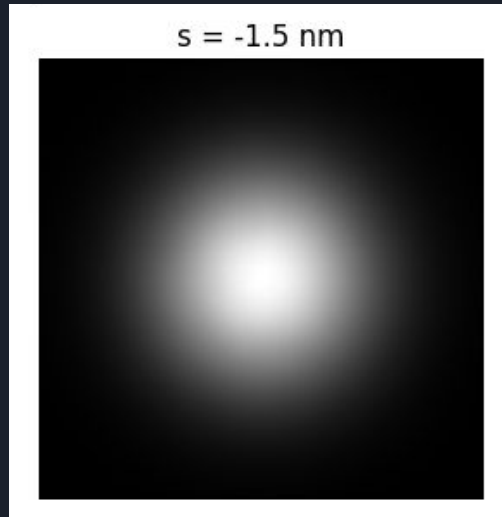


Issues Faced: Fringe Pattern was not shown

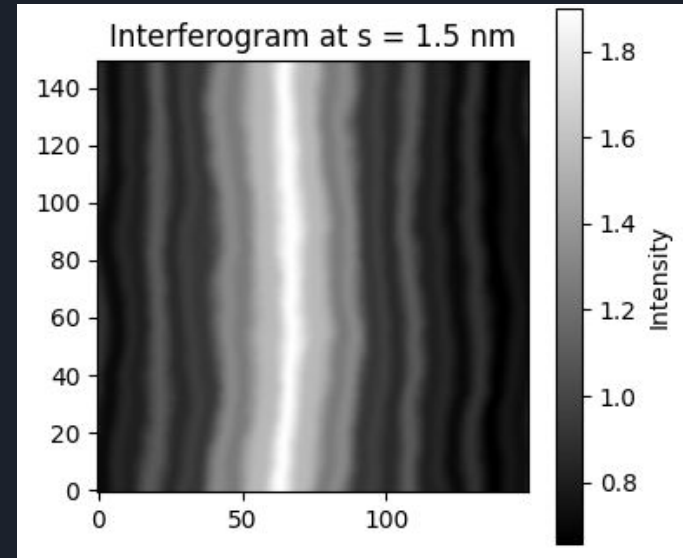
- The initial simulation without tilt produced output, but no visible spatial fringe pattern was present.
- To generate observable interference fringes, a linear phase tilt was added to the reference beam.
- After applying tilt, the interferogram displayed clear, structured fringes, making the behaviour easier to interpret and validate.
- Tilt helped confirm that the simulation was working correctly, and that interference was occurring as expected.
- The final simulation includes tilt, and it remained part of the model because it improved fringe visibility and interpretability.

Interferogram Without Tilt vs With Tilt

Without Tilt:



With Tilt:





Thank you!